IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.0

Project ID: RP26-IT-049

1. Topic (12 words max)

AR-Based Intelligent Automobile Repair Learning & Evaluation System

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	RP26-IT-049
Year	2026

5. Brief description of the research problem including references (500 words max) – references not included in word count.

Automobile education requires hands-on learning, especially when identifying engine components, performing repair tasks, and following correct safety practices. However, many students in vocational training institutes struggle with practical exposure due to limited vehicle access, lack of expert supervision, and safety concerns. Traditional lecture-based methods are not scalable and fail to provide personalized feedback.

User Persona:
Automobile engineering students, vocational trainees, and beginners in automotive repair. They often lack real-time guidance when handling engine components such as spark plugs, air filters, or batteries.

Impact:
- Social – Improves employability through better skill development.
- Technical – Reduces mechanical errors via real-time AR assistance.
- Economic – Decreases training cost and part damage.
- Ethical – Ensures unbiased AI evaluation.
- Legal – Supports adherence to mechanical safety standards.
- Environmental – Reduces waste of components used repeatedly for training.

SDG Alignment:
SDG 4 (Quality Education), SDG 8 (Decent Work), SDG 9 (Innovation), and SDG 12 (Responsible Consumption).
This problem demands a scalable training system that offers real-time guidance, feedback, safety alerts, and automated evaluation to support effective learning.

6. Brief description of the nature of the solution including a conceptual diagram (500 words max)

- The proposed AR 4.0 Automobile Learning System integrates AI, Computer Vision, Machine Learning, and Augmented Reality to support hands-on automobile training.

  Input:
  Real-time engine video feed, student actions, audio input (optional).

  Process:
  1. Vision-based component identification using CNN models.
  2. Task recognition via sequence modeling and ML classification.
  3. AR instruction system overlays arrows, labels, and guidance.
  4. Evaluation engine calculates correctness scores, identifies mistakes, and provides safety alerts.

  Output:
  • Component identification
  • Task prediction
  • AR-based instruction overlays
  • Mistake list
  • Safety warnings
  • Task completion report

  Conceptual Input → Process → Output Flow:
  Input: video, component → Process: CV + ML + AR → Output: score + guidance + safety alerts.

7. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

- *Domain Expertise:*
  Obtained through consulting automobile instructors, researching engine documentation, and studying mechanical repair procedures.

  Technical Knowledge:
  Team gains expertise in CV (OpenCV), ML models, ARCore/ARKit, Unity, and deep learning through online courses and documentation.

  Data Collection:
  Dataset of engine components, instructor reference videos, self-recorded footage, annotated using CVAT/LabelImg.

  Ethical Clearance:
  Required if collecting identifiable student video data. Optional when using synthetic or instructor-only datasets

8. Objectives and Novelty

Main Objective To develop an augmented-reality–driven intelligent learning system capable of real-time engine component identification, task recognition, practical skill verification, and automated performance evaluation for automotive education.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Senevirathne S.D – IT22279934	*Develop the vision-based engine component identification module*	*- Collect engine component dataset (images/videos)* *- Train CNN-based detection model (YOLO/MobileNet)* *- Implement real-time detection using camera feed* *- Send component ID to the next module*	*- Real-time, markerless identification of multiple engine components* *- Works under varying lighting and workshop conditions*
Jayalath S.P.S.C.K – IT22231864	Implement the student task recognition and action classification module.	- Capture action sequences (touch, rotation, disassembly) - Develop rule-based + ML action recognition model - Map interactions to tasks - Output predicted task ID	- AI-driven task recognition for engine maintenance - First attempt to classify automotive task sequences

Weerakoon Y.P.B – IT22284570	Design AR-supported task verification and guidance module.	- Develop AR overlays with step instructions - Implement visual comparison against reference actions - Highlight mistakes in real-time - Provide optional audio instructions	- Real-time AR corrective guidance - Integration of verification + instructional AR in practical learning
Leelananda M.P.D.R.I – IT21804960	Develop practical evaluation, scoring, and safety feedback module.	- Build AI scoring algorithms - Implement safety rule reasoning engine - Generate evaluation report - Provide safety alerts for high-risk components	- Automated safety-aware performance scoring - Integration of accuracy, time, and safety compliance

SLIIT UNI
THE KNOWLEDGE UNIVERSITY

9. <u>This part is to be filled by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	x	No	

b) The proposed topic exhibits novelty.

Yes	x	No	

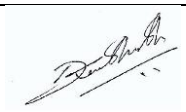
c) The student group can successfully execute the proposed project.

Yes	x	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	x	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Dr.	Dharshana	Kasthurirathna	
Co-Supervisor	Mrs	Anjalie	Gamage	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				